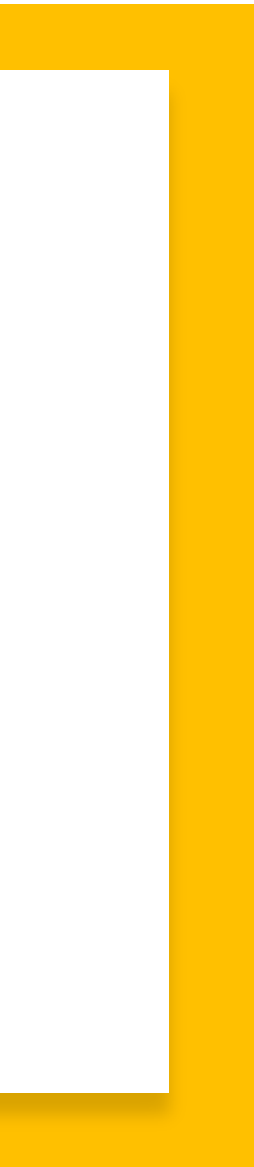
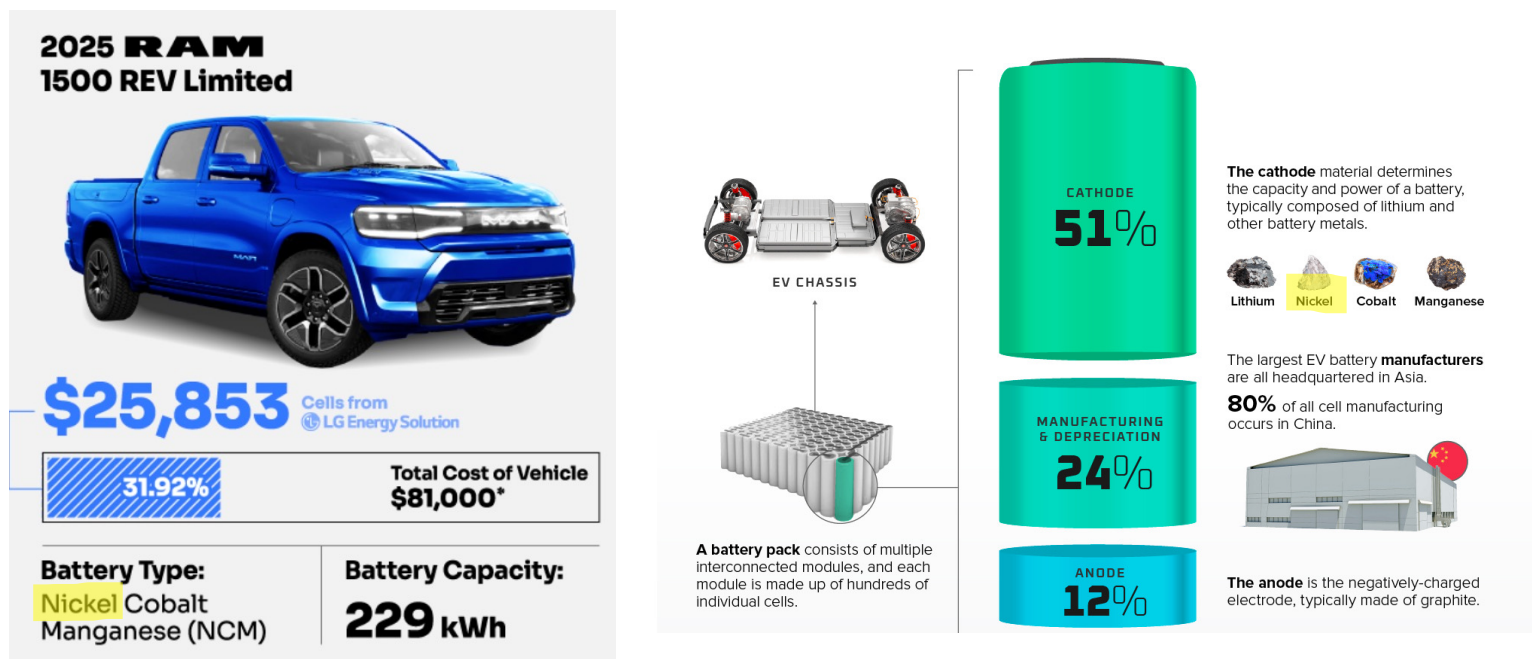




Nickel Price Prediction By LSTM



Background and Problem Definition



Battery packs make up over 30% of EV costs and rely mainly on nickel.
Rising **nickel prices** greatly cut EV profit margins.

Background and Problem Definition



an accurate nickel **price prediction model** can aid **purchasing decisions** and **cost management** strategies for manufacturers.

Background and Problem Definition

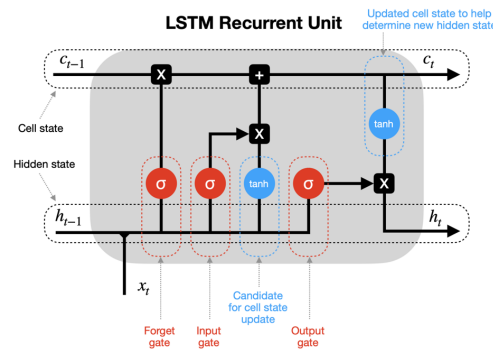


Why **Nickel**? Cobalt / Lithium / Manganese Market are new.
NOT ENOUGH DATA TO TRAIN THE MODEL.

Research Question

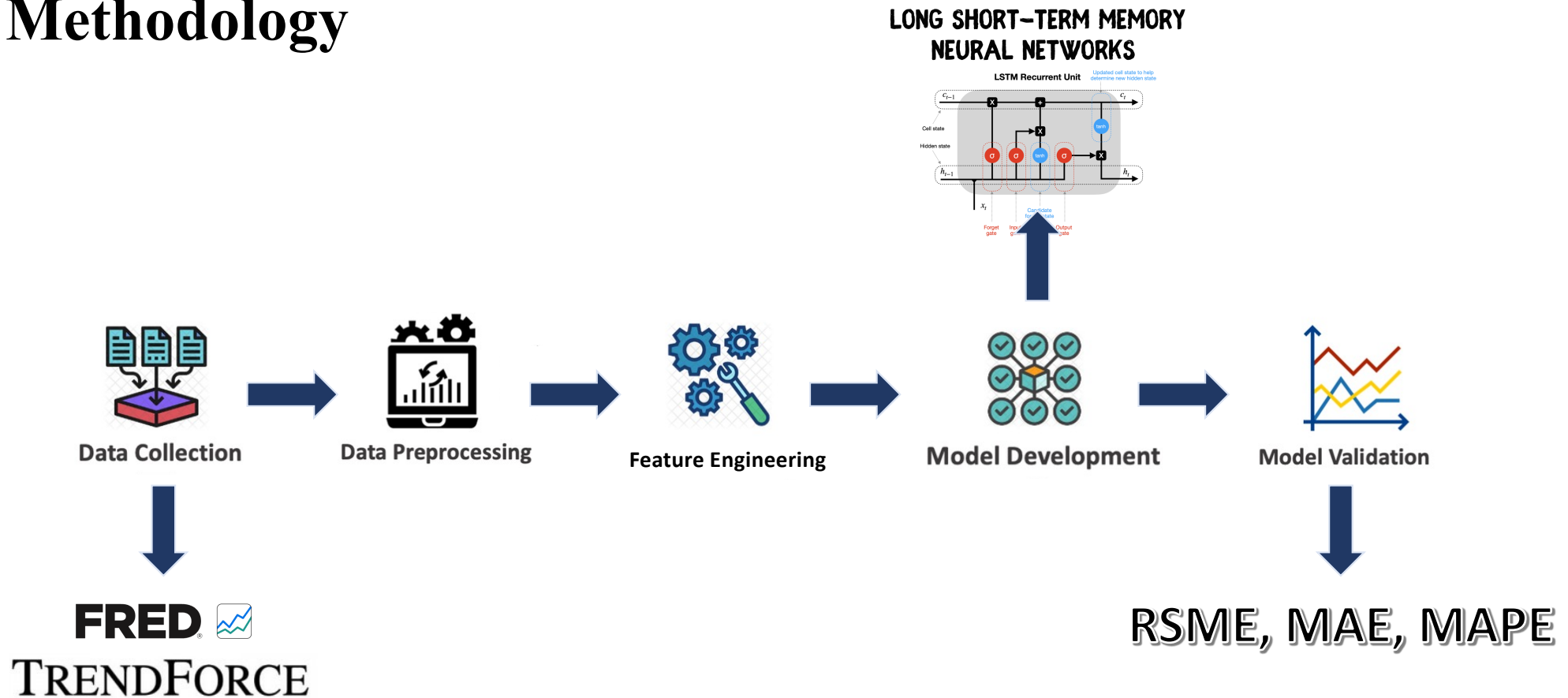
ARIMA (d, q)

LONG SHORT-TERM MEMORY NEURAL NETWORKS



Can LSTM model predict monthly nickel prices using historical economic indicators?

Methodology



Data Collection

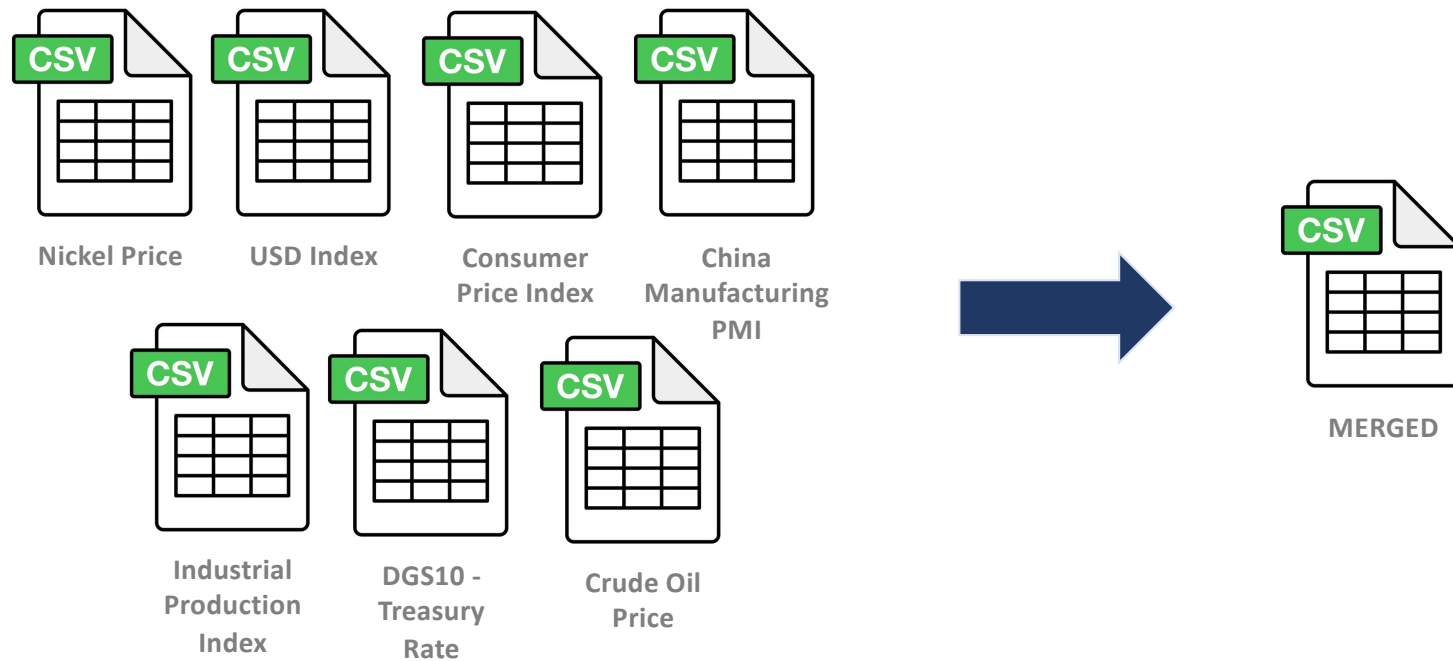


Your trusted source for economic data since 1991.



Kaggle doesn't have related dataset. Collected data from **FRED** and **TRENDFORCE**

Data Preprocessing



Collected each features individually, then **MERGED BY DATE**

Data Preprocessing



China
Manufacturing
PMI

```
# Convert to Date and set to first day of month  
chinapmi$observation_date <- as.Date(chinapmi$observation_date)  
chinapmi$observation_date <- floor_date(chinapmi$observation_date, "month")
```

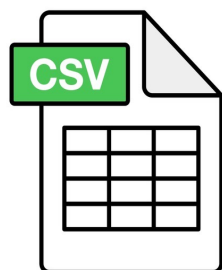


DGS10 -
Treasury
Rate

```
# Convert daily DGS10 (10-yr Treasury) to monthly average  
dgs10_monthly <- dgs10 %>%  
  mutate(observation_date = as.Date(observation_date)) %>%  
  mutate(yearmon = floor_date(observation_date, "month")) %>%  
  group_by(yearmon) %>%  
  summarise(DGS10 = mean(DGS10, na.rm = TRUE)) %>%  
  rename(observation_date = yearmon)
```

China Manufacturing PMI had different format. **DGS10** was yearly data.

Data Preprocessing



Merged

observation_date	PNICKUSDM	DTWEXBGS	CPIAUCSL
Length:135	Min. : 8298	Min. : 86.04	Min. :211.4
Class :character	1st Qu.:12828	1st Qu.: 93.67	1st Qu.:228.6
Mode :character	Median :16444	Median :111.65	Median :242.0
	Mean :17035	Mean :106.70	Mean :251.6
	3rd Qu.:19789	3rd Qu.:115.98	3rd Qu.:265.7
	Max. :33924	Max. :127.40	Max. :320.6
INDPRO	DGS10	DCOILWTICO	CHINAPMI
Min. : 84.67	Min. :0.6236	Min. : 19.72	Min. :38.80
1st Qu.: 96.54	1st Qu.:1.9346	1st Qu.: 55.52	1st Qu.:49.85
Median : 99.75	Median :2.5400	Median : 73.97	Median :50.60
Mean : 98.57	Mean :2.6410	Mean : 73.57	Mean :50.85
3rd Qu.:102.38	3rd Qu.:3.4870	3rd Qu.: 90.12	3rd Qu.:51.40
Max. :104.06	Max. :4.5391	Max. :141.06	Max. :58.40

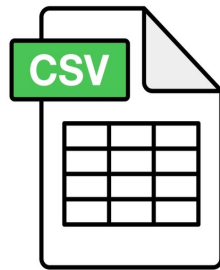
Successfully Merged. No Missing Values.

Feature Engineering – Lag Variables

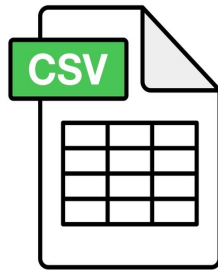


Lag Variables : **Price Momentum** (Up, Down, or Stable?)

Feature Engineering – Stationary Transformations



CHINAPMI

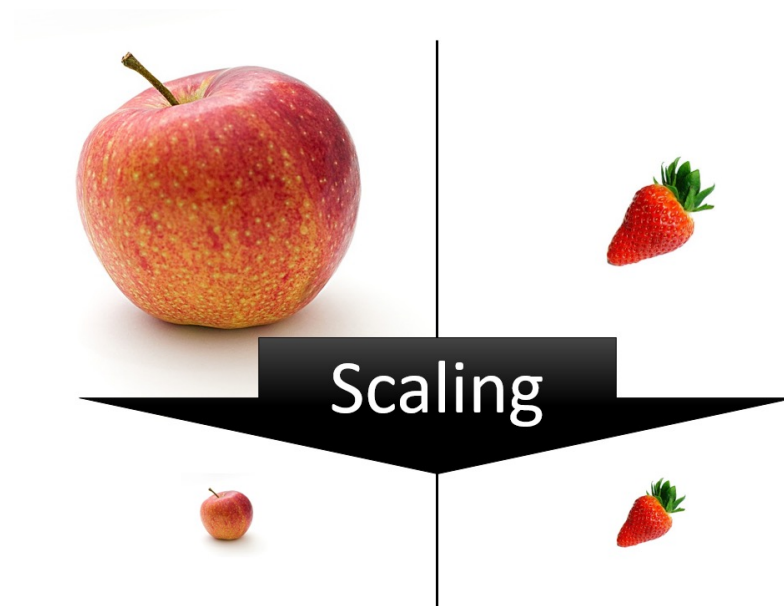


DGS10

```
# Simple differencing for interest rates  
DGS10_diff = DGS10 - dplyr::lag(DGS10, 1),  
CHINAPMI_diff = CHINAPMI - dplyr::lag(CHINAPMI, 1)
```

Differencing: **Difference between months**
Let the model **learn Changes NOT TRENDS**

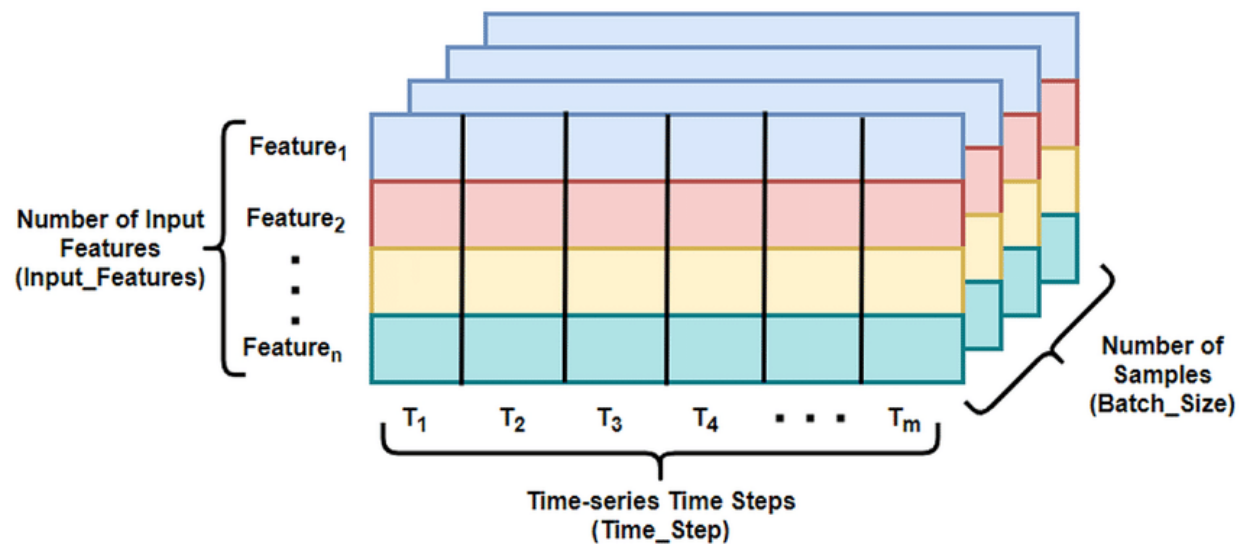
Feature Engineering – Scaling



Scaling = Standardization

Standardization: All features on same scale → Fair importance

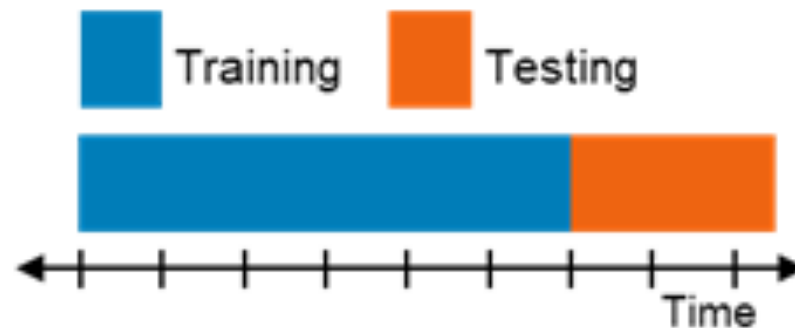
Feature Engineering – Creating Sequence



create sequences where each input consists of the past 24 months of data to predict the next month's nickel price change.

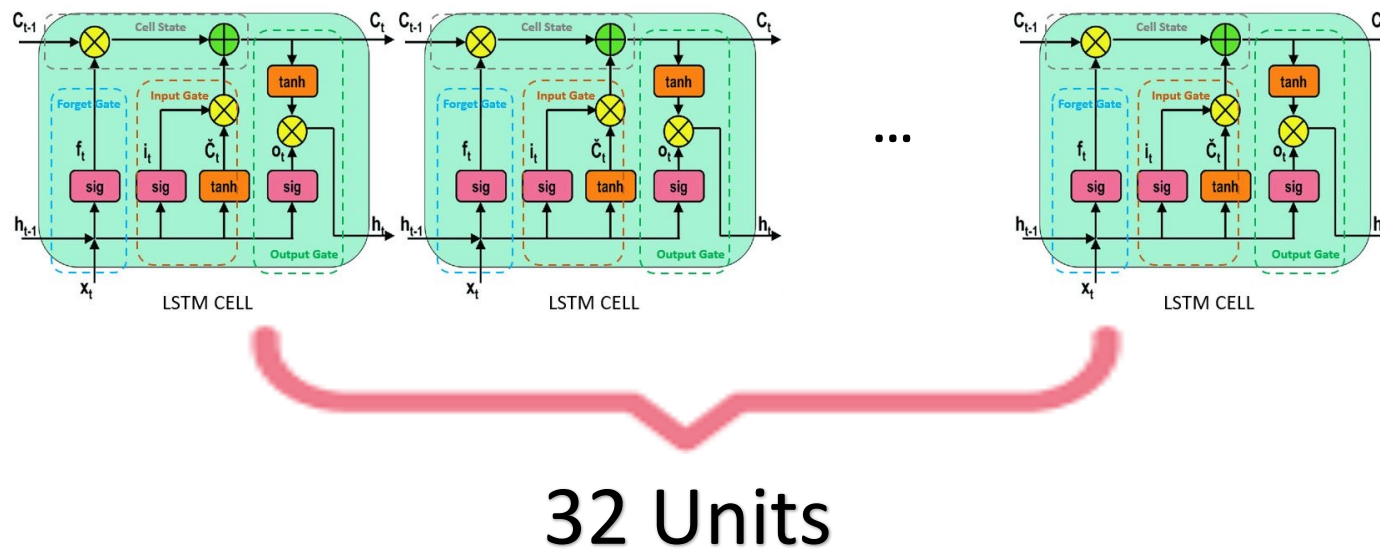
Model Development – Data Split

Time-based Estimation



First 80% of sequence = Training Set
Remaining = Testing Set

Model Development – Build



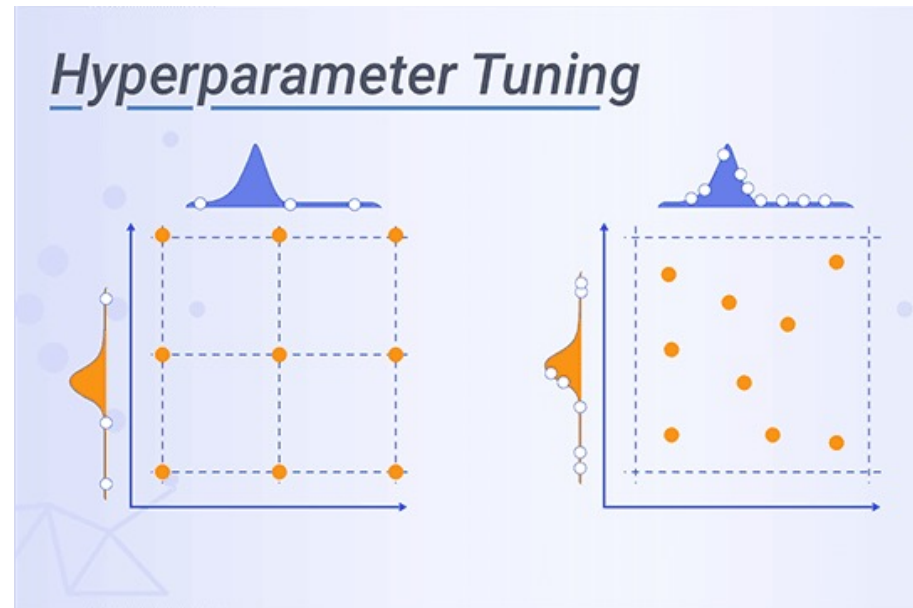
Input: 24 months $\times$ 8 features

Dropout: 20% $\rightarrow$ Prevent overfitting

LSTM: 32 units $\rightarrow$ Learn time patterns

Output: 1 value $\rightarrow$ Next month's price change

Model Development – Model Training



Max 300 epochs (stop early if no improvement) **Batch size 48** (update weights every 48 samples)
Early stopping (patience 20) → Prevent overfitting **Monitor validation loss** → Save best model

Model Validation

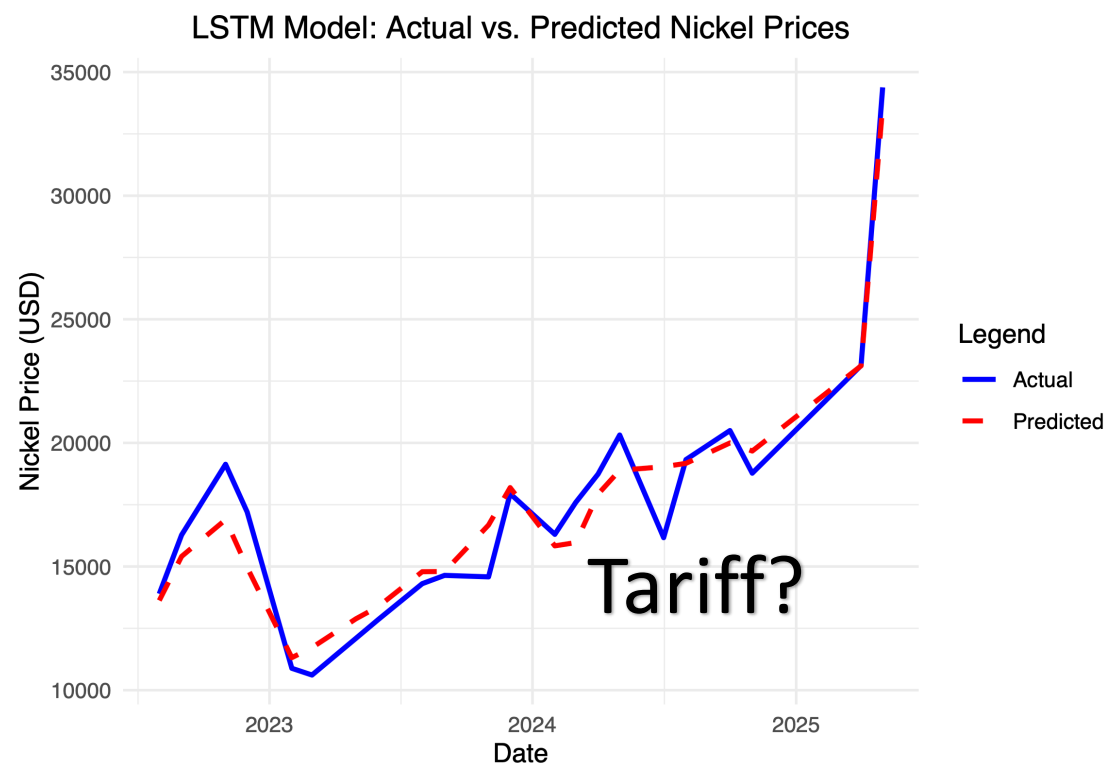
		date	actual	pred
	1	2022-08-01	13895	13625
	2	2022-09-01	16277	15409
[1] "RMSE: 1230"	3	2022-11-01	19134	16894
	4	2022-12-01	17204	14962
[1] "MAE: 953"	5	2023-02-01	10884	11315
	6	2023-03-01	10612	11723
[1] "MAPE: 5.85 %"	7	2023-05-01	12097	12887
	8	2023-06-01	12856	13380
	9	2023-08-01	14307	14791
	10	2023-09-01	14642	14800

RMSE (Root Mean Squared Error): Average prediction error in dollars

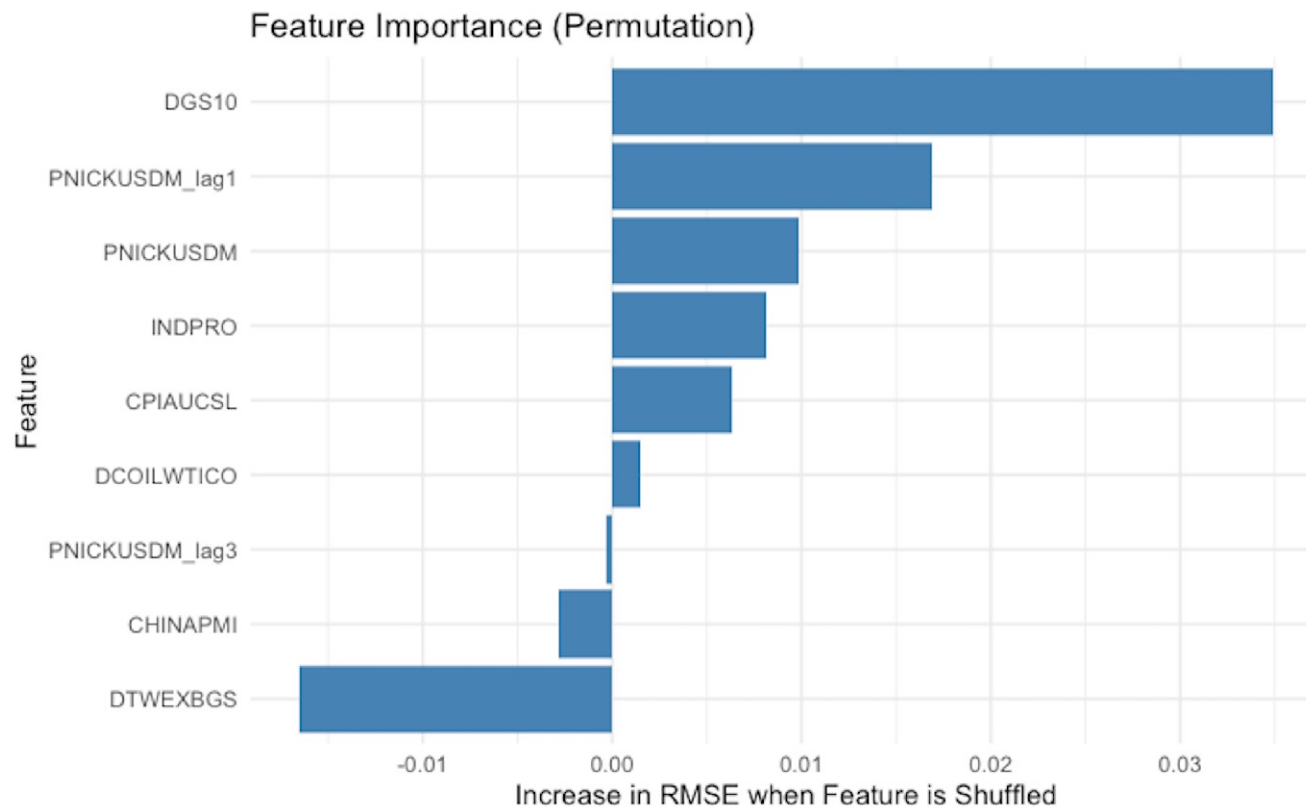
MAE (Mean Absolute Error): Average Absolute Error in dollars

MAPE (Mean Absolute Percentage Error): Average Error in %

Model Validation - Visualization



Model Validation – Feature Importance



Conclusion

Model Performance:

MAPE of 6-7% (best: 5.85%) This represents an average prediction error of approximately 6-7% relative to actual nickel prices

Limitation:

The model's performance could be affected by sudden market shocks or unprecedented events (Tariff)

This model provides a practical tool for manufacturers to anticipate nickel price movements and make informed purchasing decisions, potentially reducing cost volatility in EV production.